Training a small network

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make the trainable subset and mean objective explicit.

Model all intermediate values,retain the fixed budget,and separate reserved evaluation from structural probes and fitting.

Keep one habit: Assess the independent two-update trace using checked keys;avoid unsupported memorization or generalization labels.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Trace repeated forward loss gradient and update steps

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

The gradient describes one parameter snapshot. Mixing snapshots changes the stated update procedure.

Opening retrieval

Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a learning rate to an ordered gradient

Reference example, not the only acceptable wording: Teacher explanation: New a = 2 - 0.2 times -3 = 2.6.

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate an average rate of change over a stated interval

Reference example, not the only acceptable wording: Teacher explanation: Average slope is (0 - 4) / 2 = -2. The local slope at 2 is zero. They describe different locations or intervals.

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

Look for: Record scores and activations in dependency order

Reference example, not the only acceptable wording: Teacher explanation: z = [-3, 3], h = [0, 3], y = -5.

Pictured end goal

By the end, I can explain **training** a small network using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Updating parameters

Slopes and local change

Forward passes by hand

For the compact alternative schedule, use two 45-minute sessions to connect earlier ideas into one reproducible **training loop**. Be explicit that only the **output** weight is trained; backpropagation through all hidden parameters is not implemented here.

Keep the reserved pair outside **training** and do not oversell its evidential value.

Outcomes

Trace repeated forward loss gradient and update steps

Keep **frozen parameter**s distinct from the trained parameter

Report **training** results with their evaluation limits

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach the whole loop with a declared trainable subset

> Keep the two-row main experiment,one-pair picture and independent pair distinct.

Main configuration,complete states and frozen register

Use one hidden ReLU unit h = ReLU(x) and an **output** y = vh. The hidden **input** weight is fixed at 1, its bias is fixed at 0, and the **output** bias is fixed at 0. Only **output** weight v is trained.

This is a restricted two-layer network, not **training** every parameter of a general network.

Training pairs are (x, target) = (1, 2) and (2, 4). Both x values are positive, so hidden activations are 1 and 2. Define ordinary MSE over both rows.

Start at v = 0 and use learning rate 0.1 for exactly four full-data updates. The declared budget prevents stopping only when a preferred result appears.

Each row evaluates the parameter value after that many updates. Row zero records the starting model. The loss decreases in this specific experiment; that pattern is not a promise for all networks or rates.

Let errors be v - 2 and 2v - 4. MSE is their squared sum divided by 2, which simplifies to 2.5 times (v - 2) squared. Its gradient with respect to v is 5 times (v - 2).

At v = 0 the gradient is -10, so the first update gives v = 0 - 0.1 times -10 = 1.

The factor 2 comes from squared error; division by the number of rows makes this the gradient of the mean, not the sum. The code uses every **training** row for each update. It prints five states but performs only four updates.

All data are supplied fictional numbers.

State after this many updates · v · Predictions on fixed h[1, 2] · Errors against[2, 4] · Training MSE · Gradient for next update · Next v

> **State after this many updates:** 0
>
> **v:** 0
>
> **Predictions on fixed h[1, 2]:** [0, 0]
>
> **Errors against[2, 4]:** [−2, −4]
>
> **Training MSE:** 10
>
> **Gradient for next update:** -10
>
> **Next v:** 1

> **State after this many updates:** 1
>
> **v:** 1
>
> **Predictions on fixed h[1, 2]:** [1, 2]
>
> **Errors against[2, 4]:** [−1, −2]
>
> **Training MSE:** 2.5
>
> **Gradient for next update:** -5
>
> **Next v:** 1.5

> **State after this many updates:** 2
>
> **v:** 1.5
>
> **Predictions on fixed h[1, 2]:** [1.5, 3]
>
> **Errors against[2, 4]:** [−.5, −1]
>
> **Training MSE:** 0.625
>
> **Gradient for next update:** -2.5
>
> **Next v:** 1.75

> **State after this many updates:** 3
>
> **v:** 1.75
>
> **Predictions on fixed h[1, 2]:** [1.75, 3.5]
>
> **Errors against[2, 4]:** [−.25, −.5]
>
> **Training MSE:** 0.15625
>
> **Gradient for next update:** -1.25
>
> **Next v:** 1.875

> **State after this many updates:** 4
>
> **v:** 1.875
>
> **Predictions on fixed h[1, 2]:** [1.875, 3.75]
>
> **Errors against[2, 4]:** [−.125, −.25]
>
> **Training MSE:** 0.0390625
>
> **Gradient for next update:** Stop:four-update budget

> **Next v:** No further update

Value in this restricted network	Training status
Hidden input weight 1	Frozen
Hidden bias 0	Frozen
ReLU function	Fixed function, no fitted numerical parameter here
Output weight v	Only trainable value; start 0, end 1.875 after four updates
Output bias 0	Frozen
Data, rate.1, mean objective, four-update budget	Declared experiment configuration, not extra model parameters

Guided: **prediction**s [1.5, 3], errors [-0.5, -1], MSE 0.625, gradient -2.5, next v = 1.75. Exit: x = -1 produces 0 rather than -2 for every v.

The recorded **training** loss fell on two positive synthetic pairs; the model did not establish the rule over all real **input**s.

Complete post-freeze evaluation and structural limit

After freezing main v1.875 · h · Prediction · Target · Squared error / status

> **After freezing main v1.875:** Training row x1
>
> **h:** 1
>
> **Prediction:** 1.875
>
> **Target:** 2
>
> **Squared error / status:** 0.015625

> **After freezing main v1.875:** Training row x2
>
> **h:** 2
>
> **Prediction:** 3.75
>
> **Target:** 4

> **Squared error / status:** 0.0625

> **After freezing main v1.875:** Reserved demonstration x3
>
> **h:** 3
>
> **Prediction:** 5.625
>
> **Target:** 6
>
> **Squared error / status:** .140625; not used in the four updates

> **After freezing main v1.875:** Representation probe x−1 under proposed y2 xrule
>
> **h:** 0
>
> **Prediction:** 0
>
> **Target:** -2
>
> **Squared error / status:** 4 for any v; structural counterexample, not another training row

Separate pictured one-pair facts and scope

Train a single **output** weight while clearly marking the frozen hidden computation.

Toy h=ReLU(x), y=vh. One **training** pair x = 3, target 3; start v = 0; squared loss; rate 1/18.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

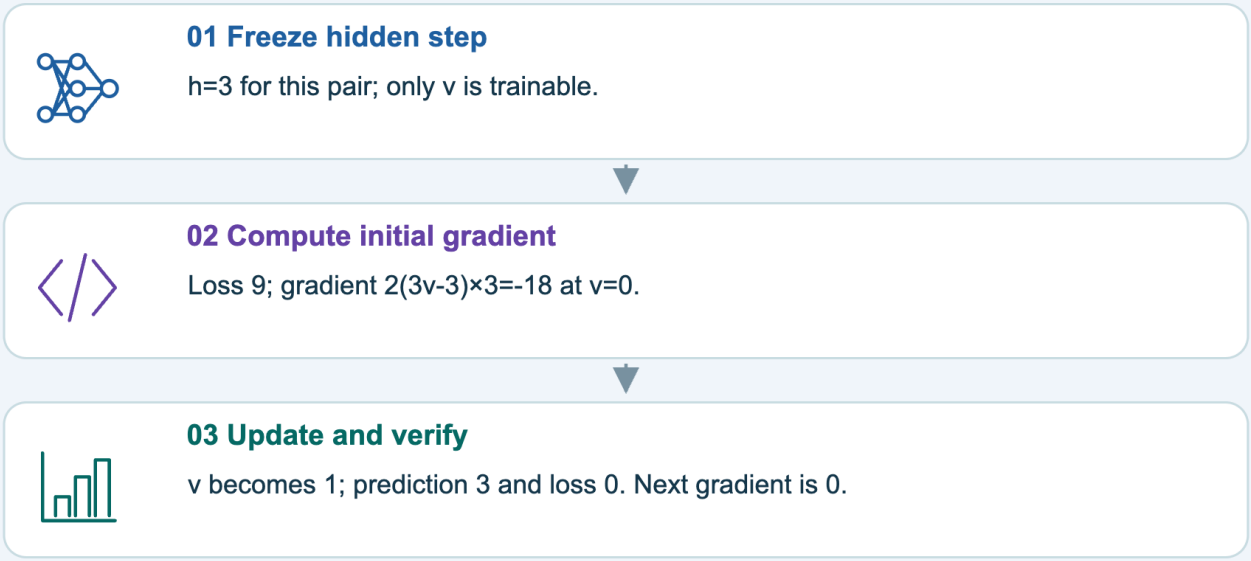
The picture is a separate one-pair experiment:x3, target 3, h3, start v0, squared error, rate 1/18. Initial gradient−18 gives v1, prediction 3, loss 0; the next gradient is0. Changing only its target to6 and restarting v0 gives initial gradient−36, v2, prediction 6, loss 0. This specially chosen rate and one pair differ from the main two-row MSE experiment.

Exact fit here does not establish memorization, learned hidden features or general image-classifier performance.

Trace fitted v and the doubled-target restart

AI ACADEMY / AI-5/12 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Freeze hidden step	h=3 for this pair; only v is trainable.
2. Compute initial gradient	Loss 9; gradient 2(3v-3)×3=-18 at v=0.
3. Update and verify	v becomes 1; prediction 3 and loss 0. Next gradient is 0.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Only the **output** weight v is being fitted in this toy network; the hidden ReLU is fixed. The **training input** gives hidden value three.

Squared-error slope points v from zero towards one, and this specially chosen rate reaches one in a single update. Zero loss on one **training** pair is exact **fitting** of that pair, not **evidence** about other **input**s.

Zero loss on one example is an exact fit to that recorded example. It does not by itself establish memorisation or a generally useful image classifier.

Change: Change target from 3 to6 and restart v = 0 with the same rate.

Prediction: First gradient=-36; v becomes 2; **prediction** 6 and loss 0.

Reason: The hidden value remains 3, but the target residual and required **output** weight double.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Training loop	A repeated sequence of prediction, loss calculation, gradient calculation and parameter update
Frozen parameter	A model value intentionally held fixed during fitting
Training budget	A declared limit on updates or other training resources
Learning curve	A record of a performance measure across training steps or time

One full-data update from v is 1

STAGE	COMPUTE WITH CURRENT V	RESULT
Forward and loss	Predictions [1, 2] for targets [2, 4]	Errors [-1, -2], MSE 2.5
Gradient	2 times (-1 times 1 - 2 times 2) / 2	Gradient -5
Update with rate 0.1	1 - 0.1 times -5	Next v is 1.5

Each row depends on the preceding calculation. Recompute predictions and loss on the next iteration; do not keep using the first gradient after parameters change.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

One full-data update from v is 1

STAGE	COMPUTE WITH CURRENT V	RESULT
Forward and loss	Predictions [1, 2] for targets [2, 4]	Errors [-1, -2], MSE 2.5
Gradient	2 times (-1 times 1 - 2 times 2) / 2	Gradient -5
Update with rate 0.1	1 - 0.1 times -5	Next v is 1.5

Each row depends on the preceding calculation. Recompute predictions and loss on the next iteration; do not keep using the first gradient after parameters change.

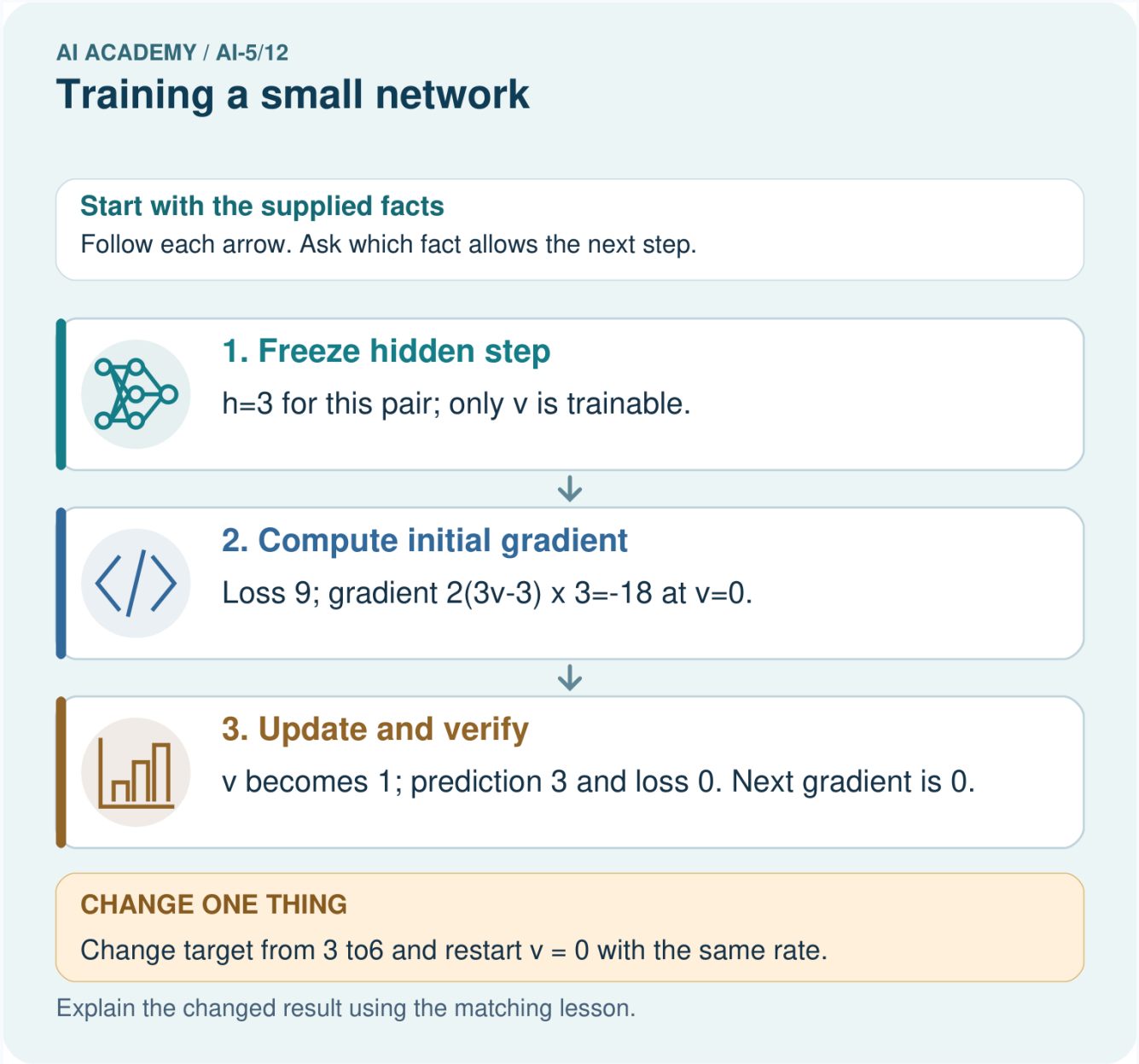
Starting from v = 1.5, calculate both training predictions, errors, MSE, gradient and the next v.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Training a small network

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: At v = 1, the sum of squared errors is 5. A learner uses gradient -10 with rate 0.1. Explain the mismatch with this lesson and give the next v under MSE.

Teacher check: Teacher explanation: The stated objective averages two rows. Its gradient is -5, giving v = 1.5. Gradient -10 corresponds to the sum objective and changes the effective update size.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Training** a small network. Proposed explanation: “Every parameter was trained”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Only v changes. Hidden weight, hidden bias and **output** bias were explicitly frozen.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Which parameter changed, and which part was not trained? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Only **output** weight v changed. The fixed ReLU hidden function h=ReLU(x) retained its declared computation. Changed-case check: First gradient=-36; v becomes 2; **prediction** 6 and loss 0.

The hidden value remains 3, but the target residual and required **output** weight double.

Changed-case reasoning: First gradient=-36; v becomes 2; **prediction** 6 and loss 0. The hidden value remains 3, but the target residual and required **output** weight double.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide the hidden activations so students can focus on errors and updates; allow a calculator for decimals. Extension: show that v - 2 is halved each step at rate 0.1, so MSE is quartered.

Explain why this is a property of this particular quadratic **training** problem.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the simultaneous update rule and identify fixed versus trained values.
7 to 18 minutes	Compute hidden activations for both training rows, then W1 and the first update.
18 to 32 minutes	Trace the next iteration with the diagram; use W2 to distinguish mean from sum.
32 to 45 minutes	Run or trace the code, marking five printed states and four actual updates.
Session 2 start to 8 minutes	Complete the guided v = 1.5 iteration and reconstruct the learning curve.
8 to 20 minutes	Freeze the final model, complete W3 and discuss the negative-input limit in W5.
20 to 35 minutes	Collect W4 independently with both iterations and the reason for no second change.
35 to 45 minutes	Use the exit response and fresh retry; save the experiment record for the next week.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main ledger and objective scaling supplied. W2 mean-gradient–5 givesv1.5; sum–10 would change effective step. Supportedv1.5 predictions1.5/3, errors–.5/–1, MSE.625, gradient–2.5, next 1.75. Independent(2, 2), v0, rate.125:initialloss 4, gradient–8; firstv 1/loss 0, nextgradient 0so secondv 1/loss 0. Frozenhidden computation still not trained. Zero loss on one pair alone is not a memorization diagnosis.

Reserved outcomes never enter the reported fitting steps.

Diagnose the current misconceptions

Every parameter was trained: Only v changes. Hidden weight, hidden bias and **output** bias were explicitly frozen.

Five printed rows mean five updates: The initial state is row zero; the loop performs four updates.

Falling **training** loss proves generalisation: Report reserved **evidence** and representation limits separately.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Keep h = ReLU(x), train only v, and use one pair (2, 2), v = 0 and rate 0.125. Use squared error and gradient 2 times (2v - 2) times 2. Perform two updates, reporting each new v and loss.

Explain why the second update changes nothing.

Teacher explanation: At start loss 4 and gradient -8. First update gives v = 1 and loss 0. The next gradient is 0, so second update keeps v = 1 with loss 0.

This is an exact fit on the one **training** pair, not proof for every **input**.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trace the first update

At v = 0, calculate both **prediction**s, errors, MSE and gradient, then update at rate 0.1.

Teacher explanation: **Prediction**s [0, 0], errors [-2, -4], MSE 10, gradient -10 and new v = 1.

W2 Check the mean convention

At v = 1, the sum of squared errors is 5. A learner uses gradient -10 with rate 0.1. Explain the mismatch with this lesson and give the next v under MSE.

Teacher explanation: The stated objective averages two rows. Its gradient is -5, giving v = 1.5. Gradient -10 corresponds to the sum objective and changes the effective update size.

W3 Freeze before checking

At final v = 1.875, calculate the **prediction** and squared error on reserved pair (3, 6). May its target be included in the four reported **training** updates?

Teacher explanation: **Prediction** 5.625, residual -0.375, squared error 0.140625. Its target must stay outside those updates if the pair is described as reserved evaluation.

W5 Expose a frozen representation limit

For the lesson architecture at x = -1, give h and **prediction** for any finite v. If target is -2, calculate squared error and explain why changing v cannot fix it.

Teacher explanation: h = 0, **prediction** 0, squared error 4. Every v is multiplied by zero. A representation or architecture change is needed for that target at this **input**.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: **prediction**s [1.5, 3], errors [-0.5, -1], MSE 0.625, gradient -2.5, next v = 1.75. Exit: x = -1 produces 0 rather than -2 for every v.

The recorded **training** loss fell on two positive synthetic pairs; the model did not establish the rule over all real **input**s.

W1 Trace the first update

W2 Check the mean convention

W3 Freeze before checking

W4 Run a new training case independently

Keep h = ReLU(x), train only v, and use one pair (2, 2), v = 0 and rate 0.125. Use squared error and gradient 2 times (2v - 2) times 2. Perform two updates, reporting each new v and loss.

Explain why the second update changes nothing.

Teacher explanation: At start loss 4 and gradient -8. First update gives v = 1 and loss 0. The next gradient is 0, so second update keeps v = 1 with loss 0.

This is an exact fit on the one **training** pair, not proof for every **input**.

W5 Expose a frozen representation limit

Independent assessment and fresh retry

W4 requires v values 1 and 1, both losses 0, and a zero second gradient. Fresh retry: one pair (1, 3), v = 1, rate 0.25. First residual -2 and gradient -4 give v = 2 with loss 1. Next gradient -2 gives v = 2.5 with loss 0.25.

Require a recomputed gradient on the second step.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **training** a small network **evidence** sheet to Image-pattern Network Investigation. Show how you can trace repeated forward loss gradient and update steps.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide the hidden activations so students can focus on errors and updates; allow a calculator for decimals. Extension: show that v - 2 is halved each step at rate 0.1, so MSE is quartered.

Explain why this is a property of this particular quadratic **training** problem.

Week 13 introduces batching and epochs with explicit update counts.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/13, ai-5/15, ai-5/18. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Trace repeated forward loss gradient and update steps

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

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>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: At v = 1, the sum of squared errors is 5. A learner uses gradient -10 with rate 0.1. Explain the mismatch with this lesson and give the next v under MSE.

Reasoned practice answer: Teacher explanation: The stated objective averages two rows. Its gradient is -5, giving v = 1.5. Gradient -10 corresponds to the sum objective and changes the effective update size.

Check every learner before discussion: Use the worked case for **Training** a small network. Proposed explanation: “Every parameter was trained”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Only v changes. Hidden weight, hidden bias and **output** bias were explicitly frozen.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change target from 3 to6 and restart v = 0 with the same rate. Require a **prediction** and a reason before comparing. Expected result: First gradient=-36; v becomes 2; **prediction** 6 and loss 0.

Reason: The hidden value remains 3, but the target residual and required **output** weight double.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

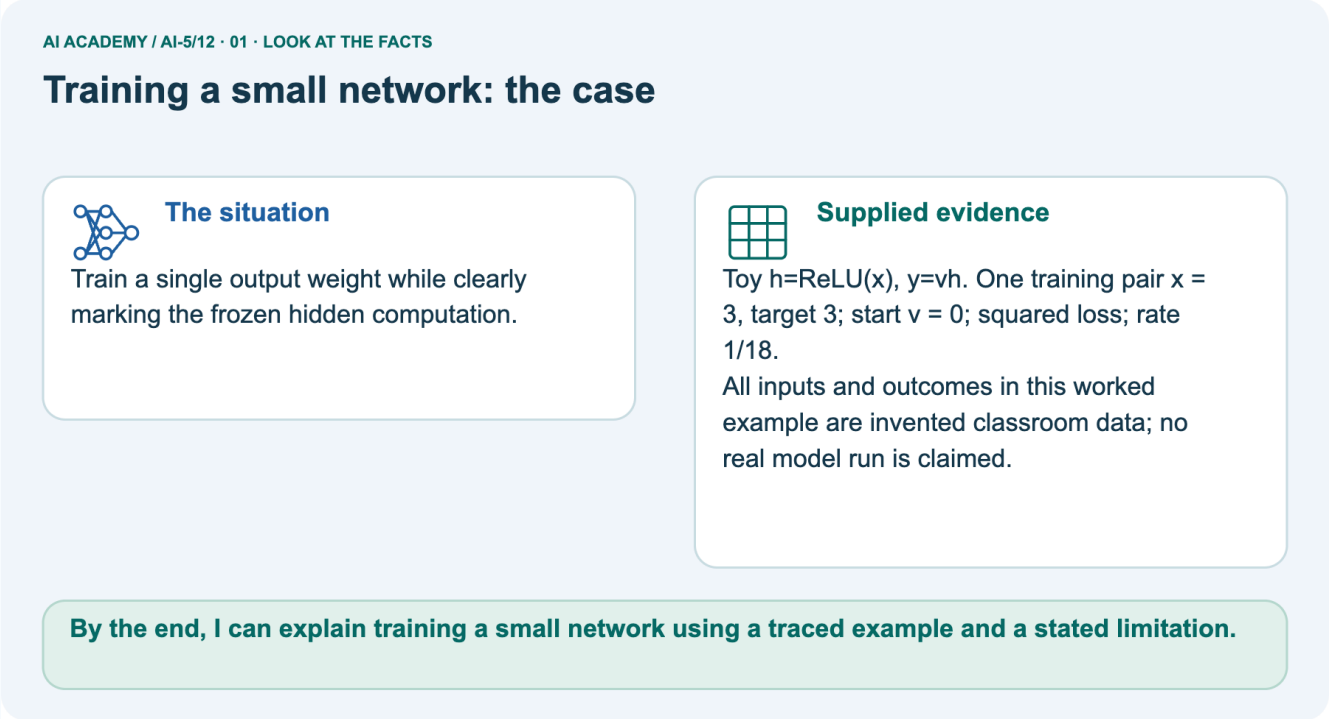
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Training a small network: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

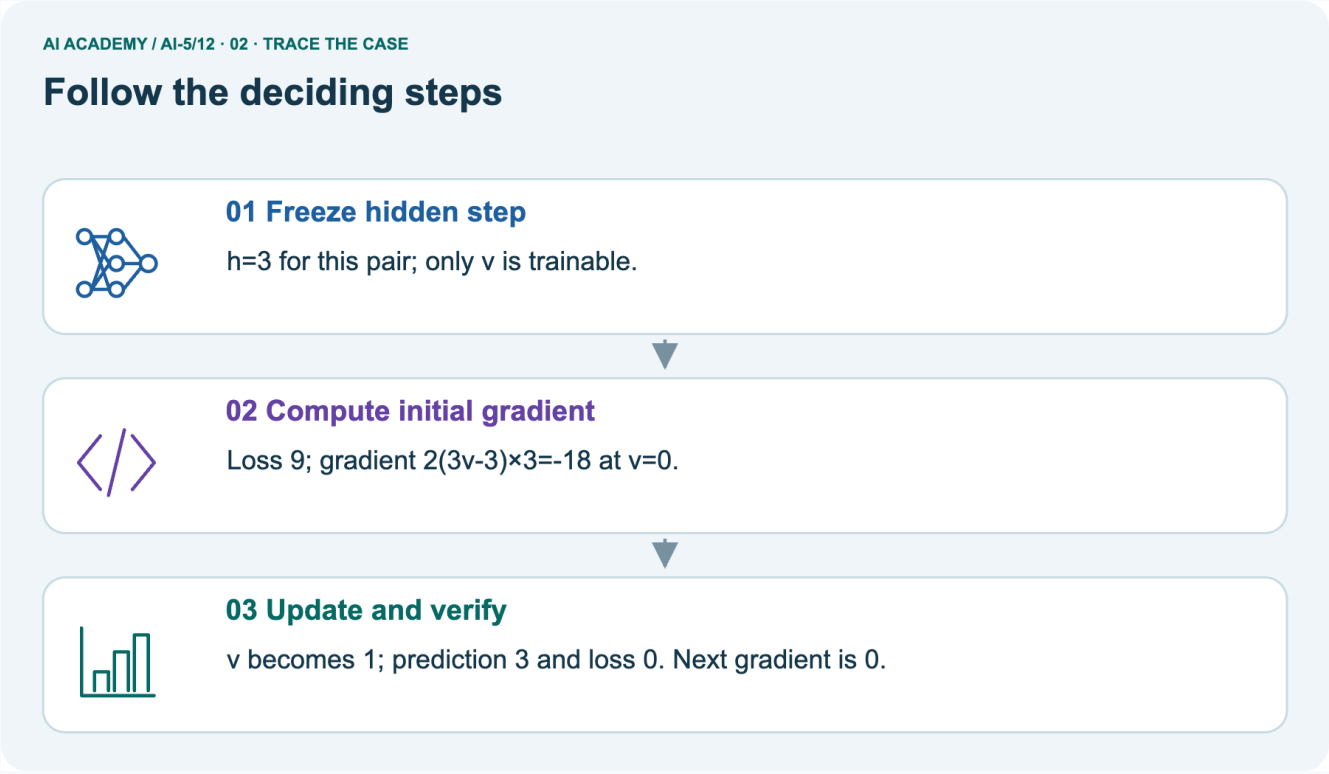
Train a single **output** weight while clearly marking the frozen hidden computation. Supplied facts: Toy h=ReLU(x), y=vh. One **training** pair x = 3, target 3; start v = 0; squared loss; rate 1/18.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



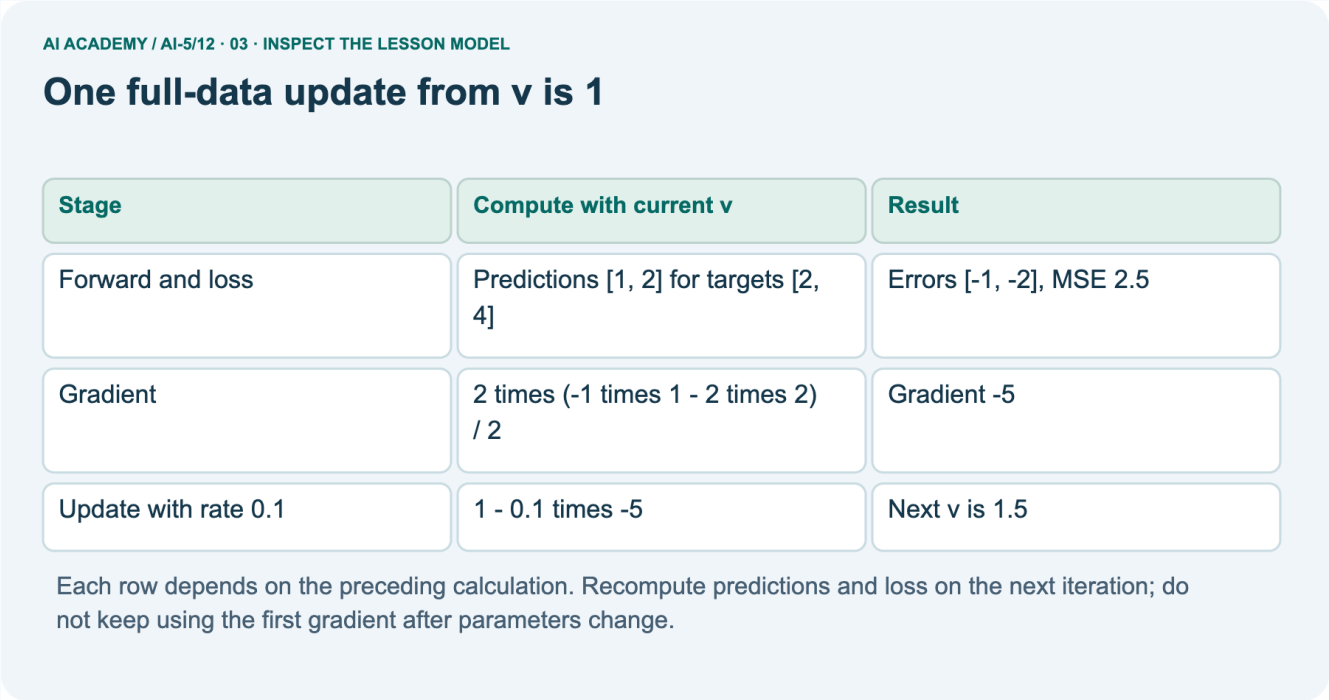
Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze hidden step: h=3 for this pair; only v is trainable. → Compute initial gradient: Loss 9; gradient 2(3v-3)×3=-18 at v=0. → Update and verify: v becomes 1; **prediction** 3 and loss 0. Next gradient is 0.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. One full-data update from v is 1



AI ACADEMY / AI-5/12 · 03 · INSPECT THE LESSON MODEL

One full-data update from v is 1

Stage	Compute with current v	Result
Forward and loss	Predictions [1, 2] for targets [2, 4]	Errors [-1, -2], MSE 2.5
Gradient	2 times (-1 times 1 - 2 times 2) / 2	Gradient -5
Update with rate 0.1	1 - 0.1 times -5	Next v is 1.5

Each row depends on the preceding calculation. Recompute predictions and loss on the next iteration; do not keep using the first gradient after parameters change.

Follow the labelled picture with the matching explanation. Open the vector diagram

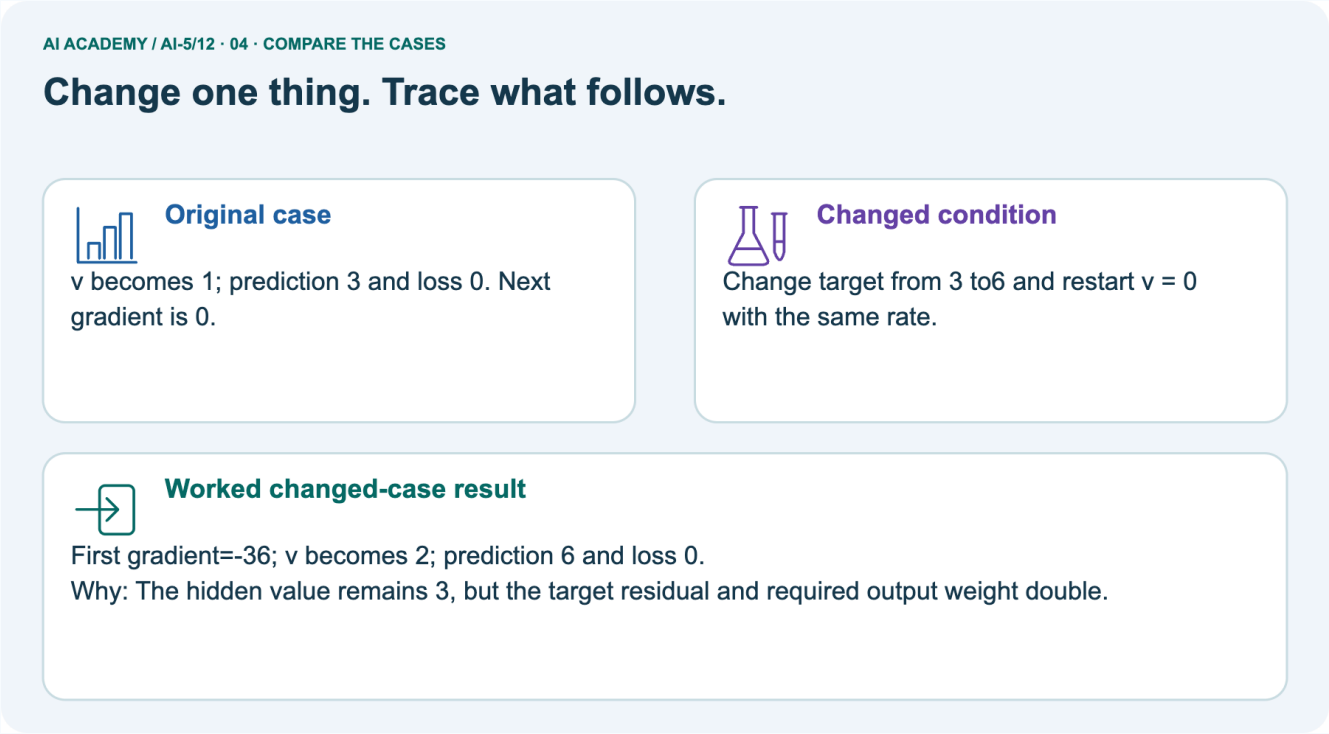
One full-data update from v is 1. Forward and loss | **Prediction**s [1, 2] for targets [2, 4] | Errors [-1, -2], MSE 2.5. Gradient | 2 times (-1 times 1 - 2 times 2) / 2 | Gradient -5.

Update with rate 0.1 | 1 - 0.1 times -5 | Next v is 1.5

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: v becomes 1; **prediction** 3 and loss 0. Next gradient is 0. Changed condition: Change target from 3 to6 and restart v = 0 with the same rate.

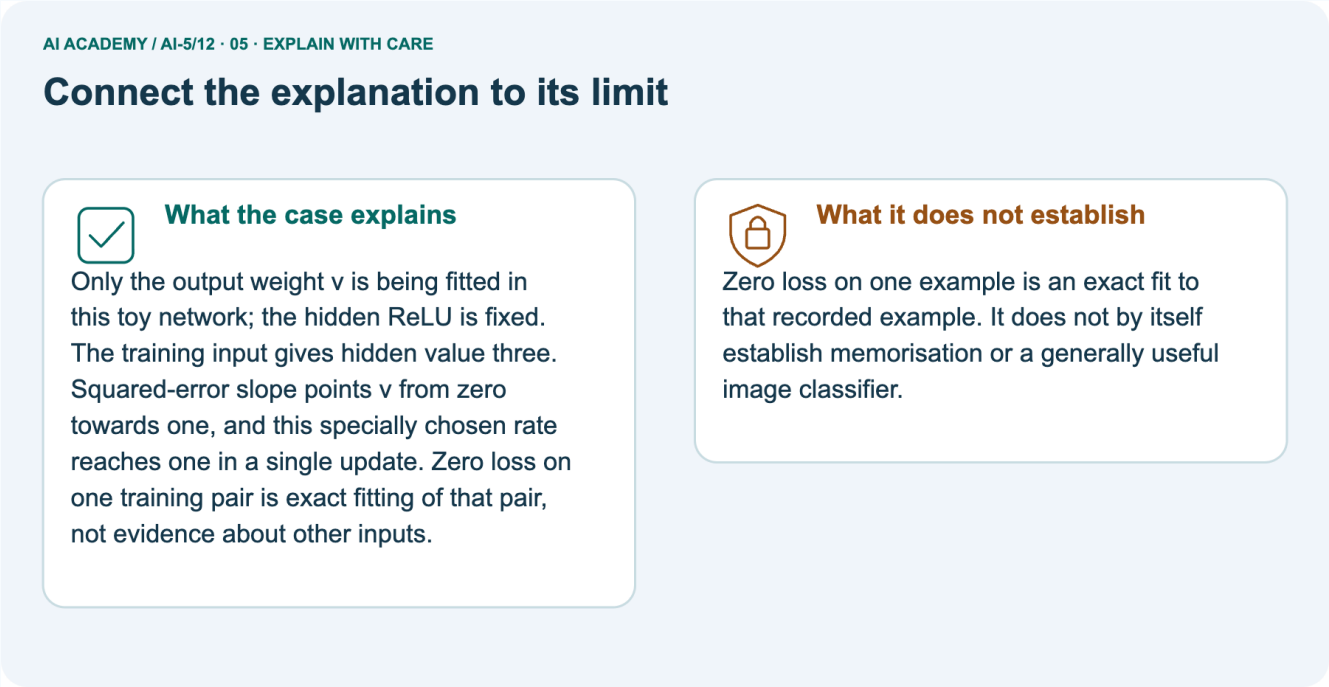
Expected result: First gradient=-36; v becomes 2; **prediction** 6 and loss 0. Reason: The hidden value remains 3, but the target residual and required **output** weight double.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The hidden value remains 3, but the target residual and required **output** weight double.

05. Connect the explanation to its limit



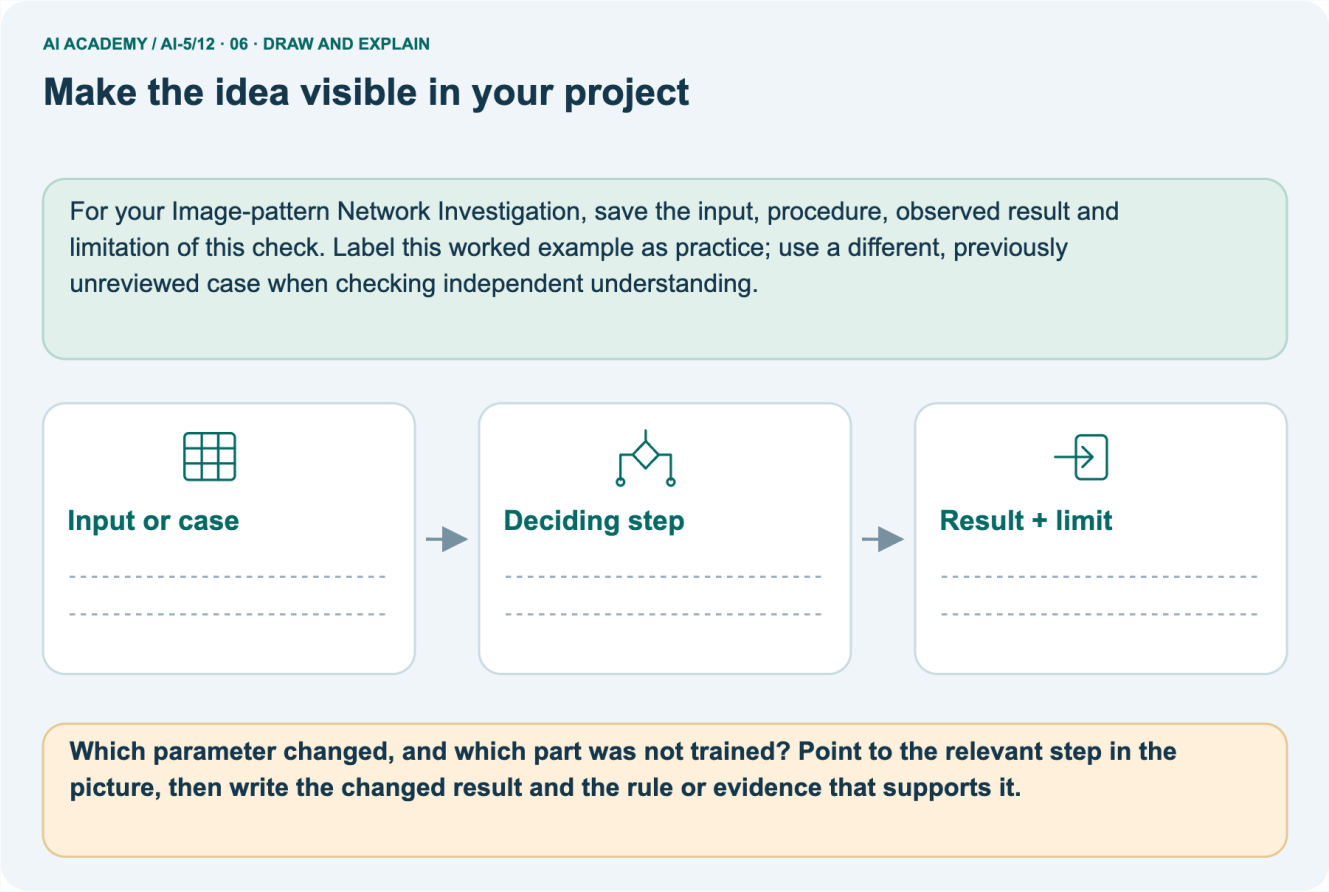
Follow the labelled picture with the matching explanation. Open the vector diagram

Only the **output** weight v is being fitted in this toy network; the hidden ReLU is fixed. The **training input** gives hidden value three. Squared-error slope points v from zero towards one, and this specially chosen rate reaches one in a single update. Zero loss on one **training** pair is exact **fitting** of that pair, not **evidence** about other **input**s. Limit: Zero loss on one example is an exact fit to that recorded example. It does not by itself establish memorisation or a generally useful image classifier.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see. Keep the stated scope: Zero loss on one example is an exact fit to that recorded example. It does not by itself establish memorisation or a generally useful image classifier.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which parameter changed, and which part was not trained? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Only **output** weight v changed. The fixed ReLU hidden function h=ReLU(x) retained its declared computation. Changed-case check: First gradient=-36; v becomes 2; **prediction** 6 and loss 0.

The hidden value remains 3, but the target residual and required **output** weight double.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Training a small network: the case

02. Follow the deciding steps

03. One full-data update from v is 1

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.